RBE 595 - HW 7

Real-Robot Implementation with Multi-View Eye-in-Hand Point Clouds

Fall 2026

This assignment should be completed with your project team - 100 points

In your HW 6, you developed the grasping pipeline in simulation using two fixed RGB-D cameras. In this assignment, you will transfer the same pipeline to the real robot. The main difference is that the real-robot setup uses a wrist-mounted RGB-D camera (eye-in-hand) rather than two fixed cameras.

Because the camera moves with the robot, you will observe the target object from at least two different viewpoints. At each viewpoint, acquire a point cloud, transform it into a common reference frame, and combine the observations to generate an augmented multi-view point cloud. You will then apply the same preprocessing, object segmentation, grasp-synthesis, grasp-orientation, and grasp-execution pipeline developed in HW 6.

Overall workflow

- Move the eye-in-hand camera to two or more observation viewpoints.
- Acquire a point cloud from each viewpoint.
- Transform every point cloud into the robot base frame.
- Merge and downsample the clouds to create an augmented point cloud.
- Filter, segment, and cluster the objects.
- Synthesize a grasp and determine the approach direction.
- Move the real robot to the pre-grasp/grasp poses and execute the grasp.

Important notes

- Write your implementation as parametrically and modularly as possible. You should be able to change observation poses, workspace bounds, filter settings, and grasp parameters without rewriting the pipeline.
- The point clouds acquired from different viewpoints cannot be merged directly in their camera frames. Each cloud must first be transformed into the same reference frame, preferably the robot base frame.
- Use at least two meaningfully different viewpoints. Additional viewpoints may be used if portions of the target object remain self-occluded. These can be predetermined and hardcoded viewpoints.
- Use conservative robot motions while testing observation poses and grasp motions. Verify that the camera, gripper, robot links, object, and table remain clear of unintended collisions.

Deliverable:

- Submit a report that briefly explains your implementation steps. At the end of each stage take a screenshot of your visualized point cloud and include it to your report.
- Submit a video of the robot executing the task.
- There will be a demo day